

Response to W2 of Reviewer Q8cZ: Empirical Evidence of Strict-Saddle Behavior Along PROBE Trajectories

Experimental setup

We run PROBE on the LLM data-curation task using five independent trajectories. Following the implementation setting in the appendix, we use 100 outer iterations, at most 200 inner-loop steps, five conjugate-gradient steps, batch size 32, LoRA rank 8, and $\mu = 0.1$. All other settings follow the appendix. At every perturbation-triggered lower-level stationary-point candidate, we additionally record the local curvature.

Results

Among 64 stationary-point candidates encountered by PROBE, 61 were numerically identified as strict saddles, none was identified as a non-strict saddle, and three remained unresolved.

Thus, all 61 candidates that could be conclusively classified exhibited strict-saddle behavior.

Trajectory	Candidates	Strict	Non-strict	Unresolved
1	8	8	0	0
2	20	19	0	1
3	7	7	0	0
4	11	10	0	1
5	18	17	0	1
Total	64	61	0	3

Table 1: Numerical classification of the perturbation-triggered lower-level stationary-point candidates encountered along five PROBE trajectories. **Unresolved** refers to the points that cannot be classified numerically.

At each perturbation trigger, we applied two numerical criteria: a **dimension-normalized gradient-norm criterion** for approximate stationarity and a **random-start Lanczos criterion with residual-based error control** for detecting negative Hessian curvature (Kuczyński and Woźniakowski, 1992; Stewart, 1991). A point was treated as an approximate stationary point when its lower-level normalized gradient norm was at most $\rho = 10^{-3}$. We then used two *independently* initialized Lanczos runs, with up to 30 steps each, to estimate the minimum Hessian eigenvalue. A candidate was classified as a strict saddle when $\hat{\lambda}_{\min} + r < -\gamma$, where r is the Lanczos residual and $\gamma = 10^{-6}$. Of the 64 candidates, 61 satisfied this strict-saddle criterion. The remaining three produced negative estimates in both Lanczos runs but could not be conclusively classified because of their numerical residuals; they were therefore marked as unresolved. Thus, all conclusively classified candidates were strict saddles, and no non-strict saddle was detected.

Summary

The results show consistent negative-curvature behavior at the lower-level saddle candidates encountered by PROBE, providing direct numerical support for the strict-saddle assumption for our paper.

References

- Kuczyński, J. and Woźniakowski, H. (1992). Estimating the largest eigenvalue by the power and lanczos algorithms with a random start. *SIAM Journal on Matrix Analysis and Applications*, 13(4):1094–1122.
- Stewart, G. W. (1991). Two simple residual bounds for the eigenvalues of a hermitian matrix. *SIAM Journal on Matrix Analysis and Applications*, 12(2):205–208.